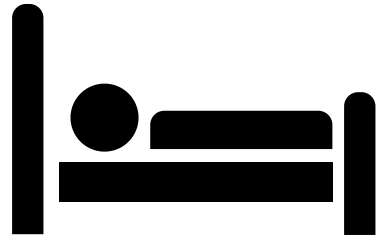


ICU Capacity Optimization

Data-Driven Insights for Critical Care
Resource Management

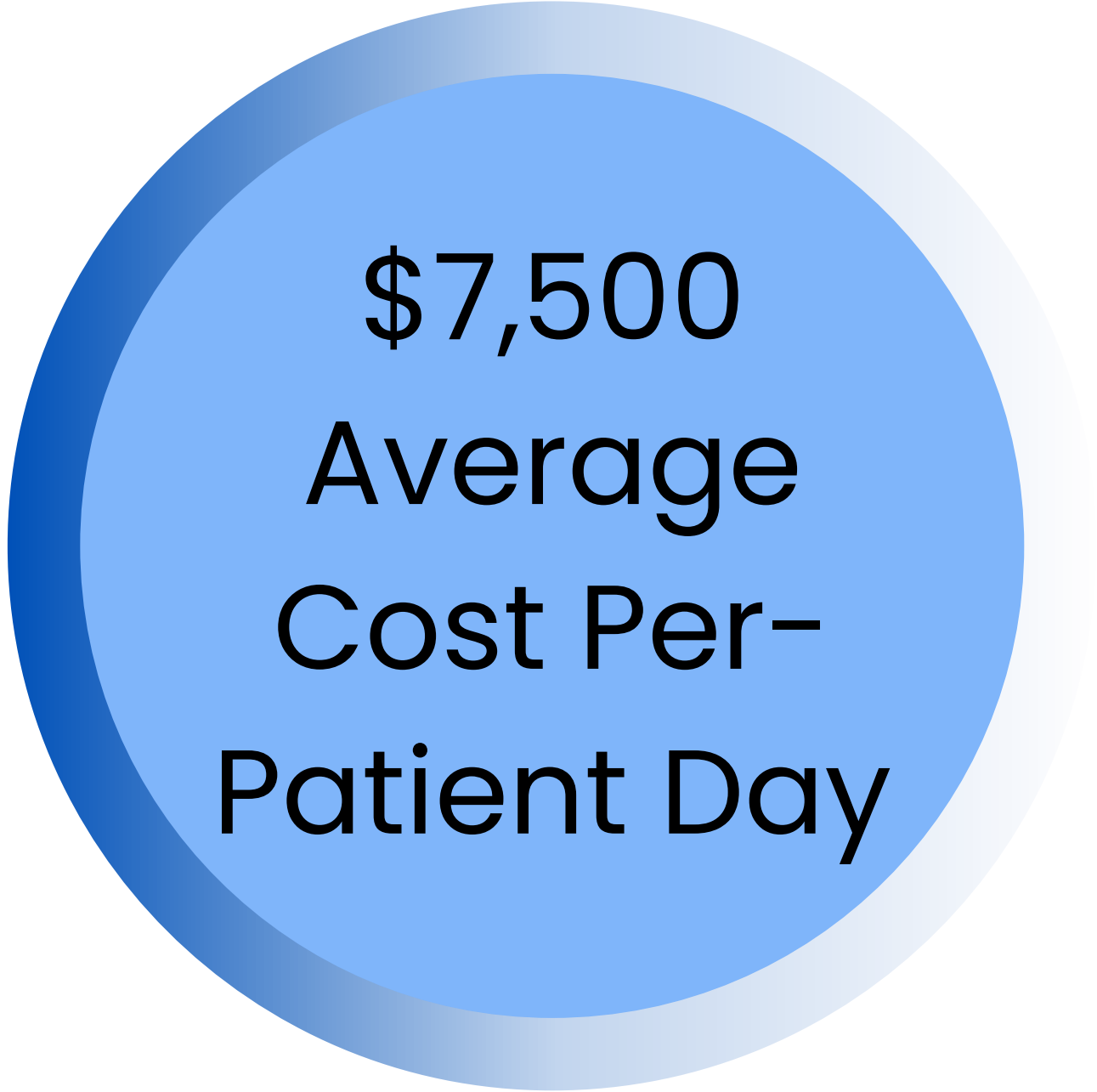
Company: CriticalPulse Insights

The Critical Care Crisis



ICU beds: the most expensive and scarce hospital resource.

- \$5,000–\$10,000+ per patient per day
- Strategic optimization = millions in potential savings.



\$7,500
Average
Cost Per-
Patient Day

Our Analysis



140 ICU Admissions

100 Unique Patients



9 ICU Types

Specialized Units Analyzed

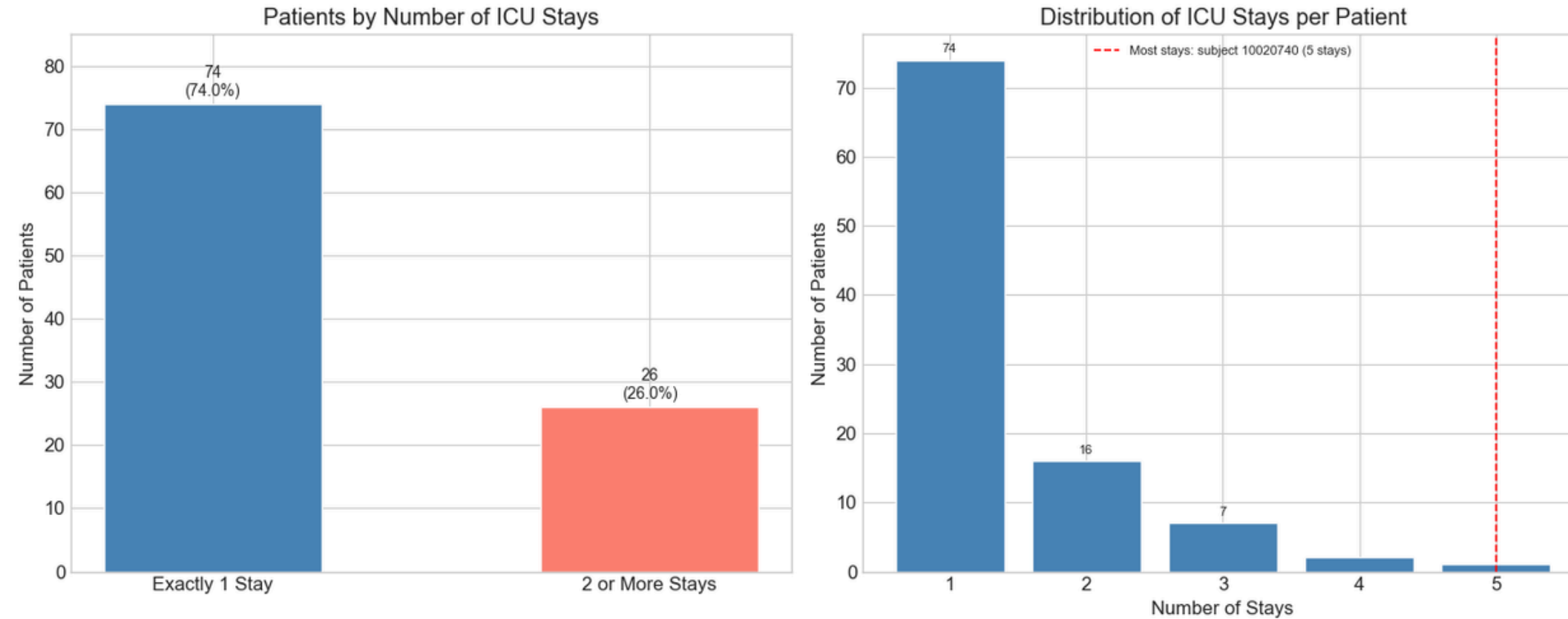


3.7 Days Avg Stay

2.2 Day Median

✓ Zero. missing values | ✓ Zero duplicate stays | ✓ High data quality

ICU Stay Summary | 26.0% of patients had multiple stays | Max: 5 stays (subject 10020740)



Patient Journey

26% of patients had multiple ICU stays

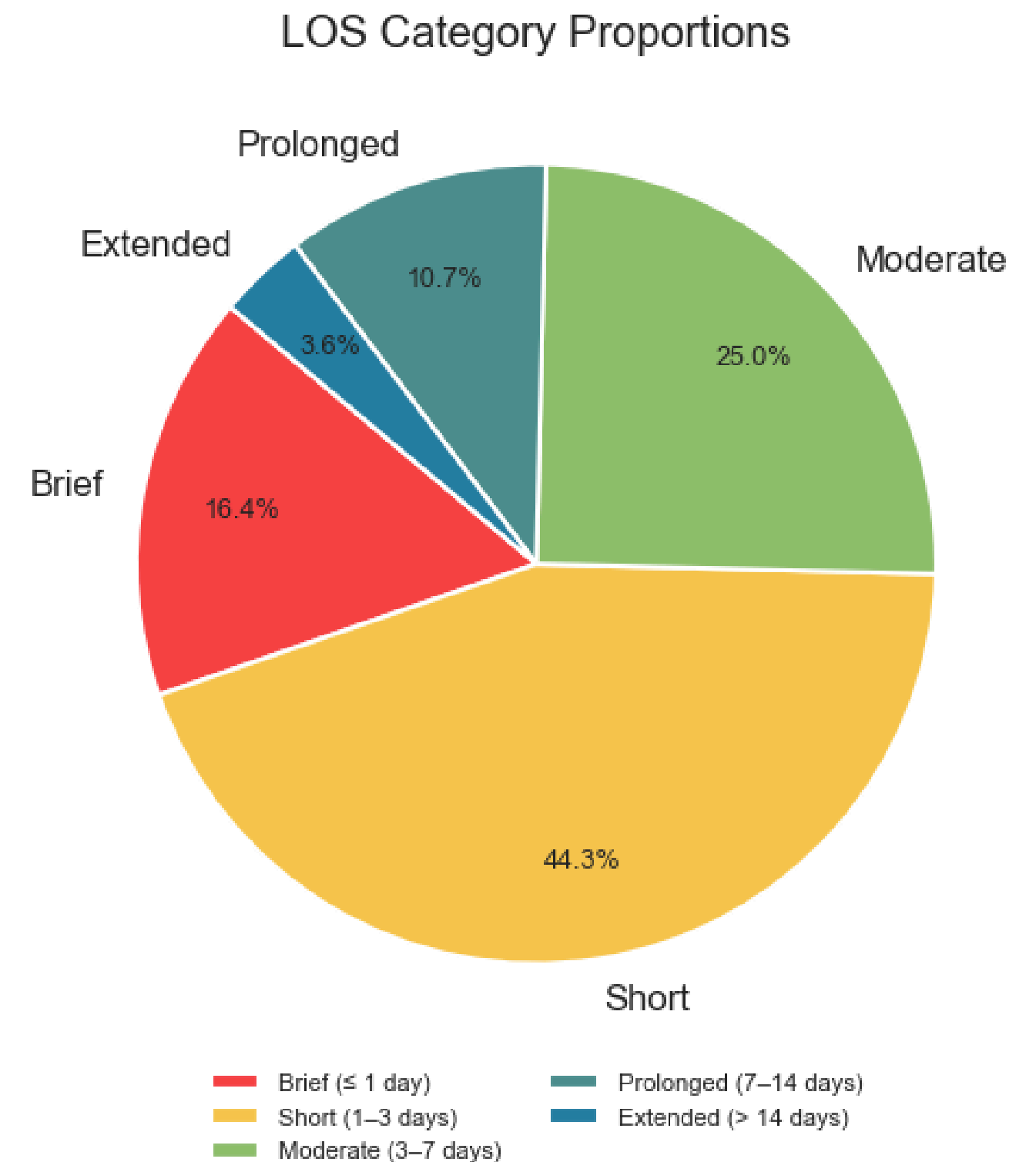
40 out of 140 required more than one ICU admission

What this Means?

- ✓ Chronic conditions requiring care
- ✓ Post-surgical complications
- ✓ Transfer patterns between specialties

The Majority Complete Care Quickly

- **Brief and Short stays (60.7%)**
 - post-surgical monitoring, observation after cardiac events, or rapid stabilization of acute conditions.
- **Moderate stays (25.0%)**
 - typically involve complications, slower-than-expected recovery
- **Prolonged and Extended stays (14.3%)**
 - complex cases: multi-organ failure, severe trauma, neurological injuries



Complexity Definitions (Method)

✓ Complexity signals:

- Repeat ICU stays:
 - 1 vs 2+ stays per patient
- Transfers:
 - first_careunit $\neq$ last_careunit
- LOS threshold: 3 days

✓ Simple

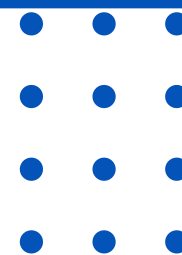
- Single stay
- No transfer
- LOS $\leq$ 3 days

✓ Moderate

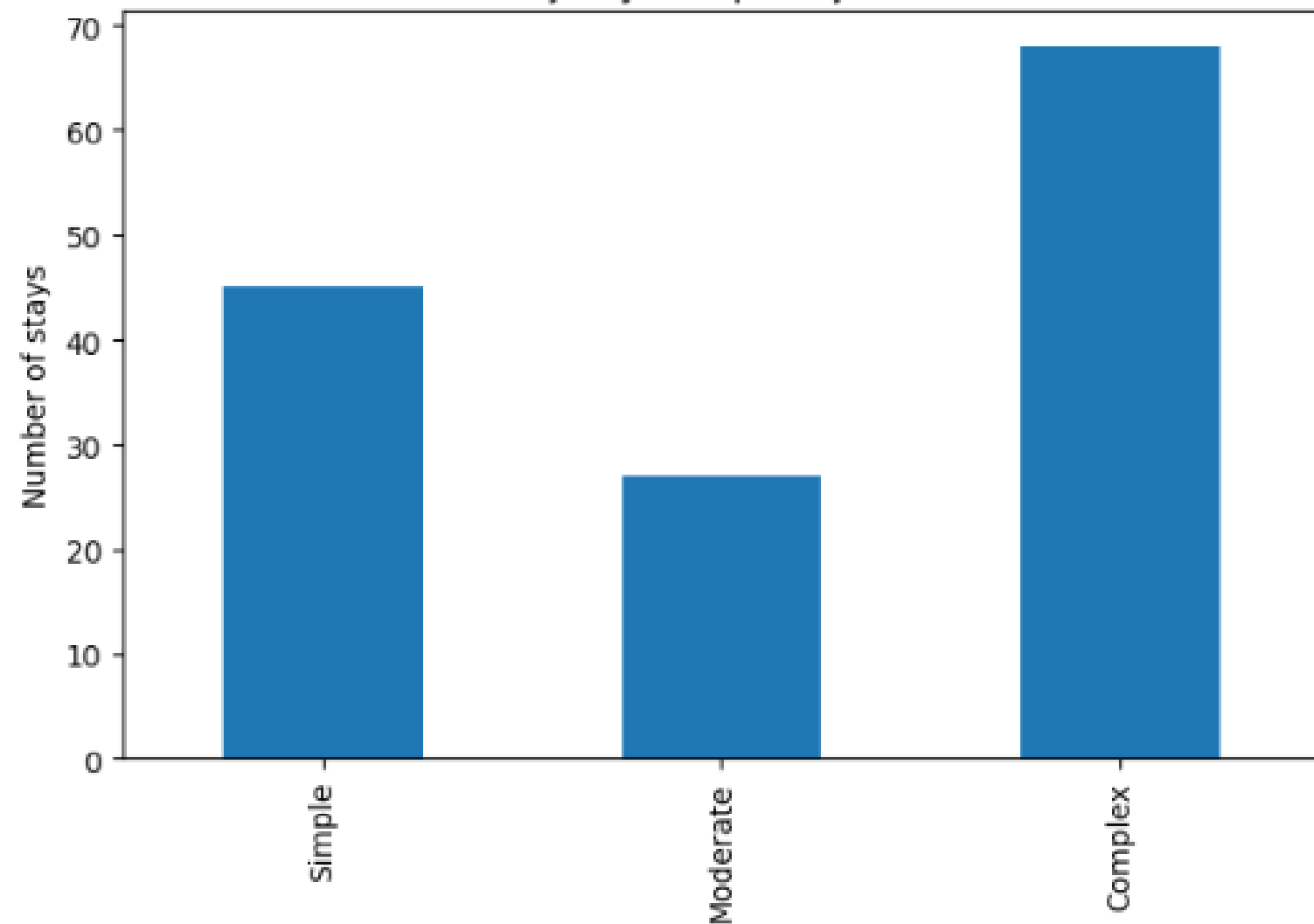
- Single stay
- Exactly one of:
 - Transfer **OR**
 - LOS $>$ 3 days

✓ Complex

- 2+ ICU stays (repeat patient)
OR
- Transfer AND LOS $>$ 3 days



Stays by Complexity Level



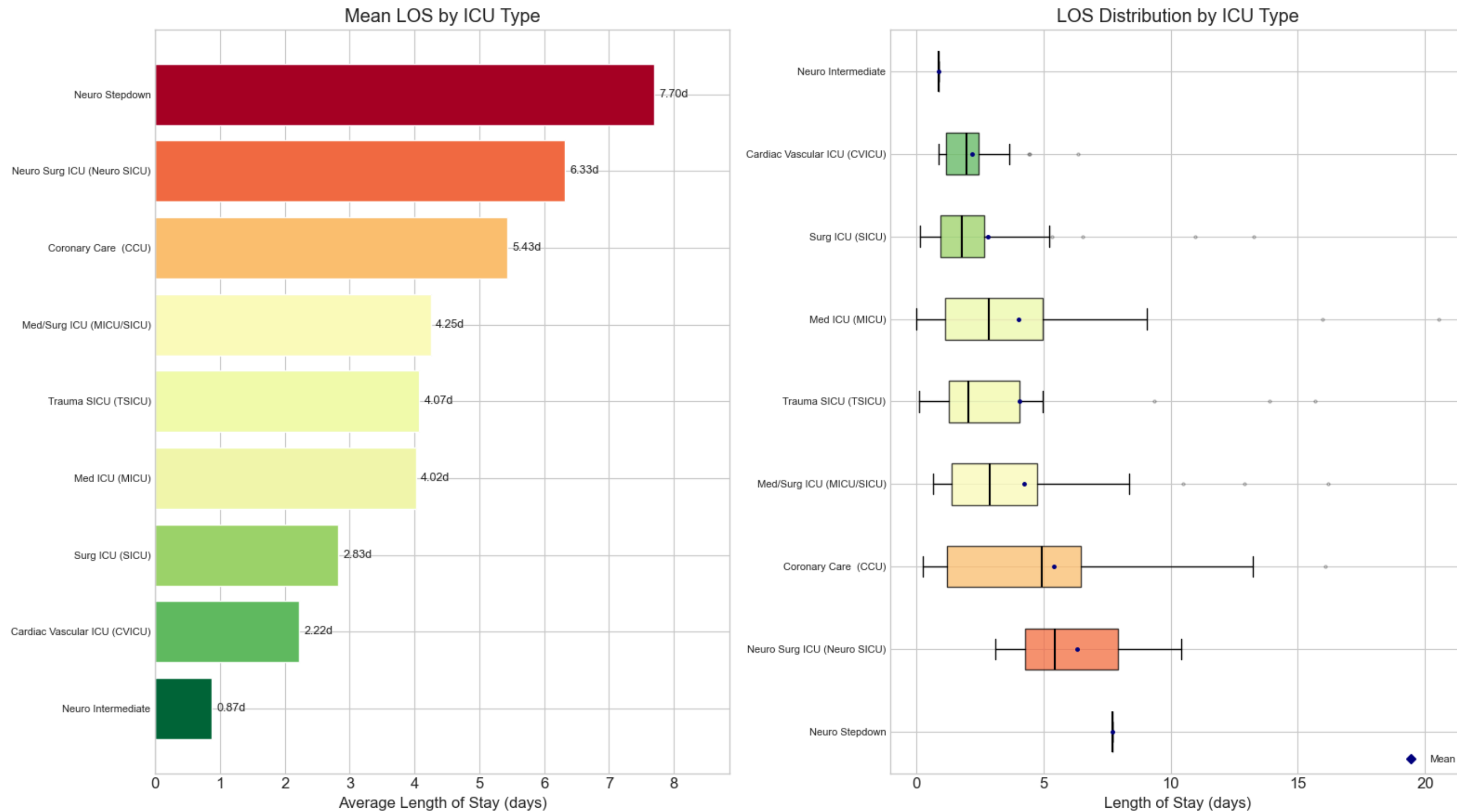
Complexity Mix + LOS by Complexity (Results)

- **Complex accounts for 48.6% of stays (68 / 140)**
- **Mean LOS (days): Simple 1.44 | Moderate 6.23 | Complex 4.15**
- **Median LOS (days): Simple 1.31 | Moderate 5.09 | Complex 2.40**

What this Means?

- ✓ **Moderate captures long single-stay utilization (highest average LOS)**
- ✓ **Complex captures repeat utilization + cross-unit transitions**
- ✓ **Complexity isn't only LOS, it's repeat demand + transitions**

Average Length of Stay by ICU Type



Staffing Recommendations by ICU Specialty

Medical ICU (Highest Resource



Demand)

- 214 total ICU days (highest)
- Largest sustained staffing need
- Requires flexible staffing for unpredictable cases

Neurological ICU (Longest LOS)



- 5.5 day average LOS (longest)
- Requires specialized neuro-trained staff
- Longer monitoring protocols

Surgical ICU (High Throughput Model



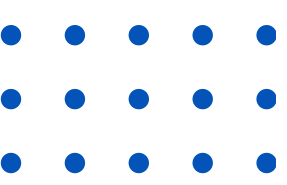
- Shorter LOS (3.3 days avg)
- Predictable recovery patterns
- Focus on efficient bed turnover

Average LOS by ICU Specialty Category

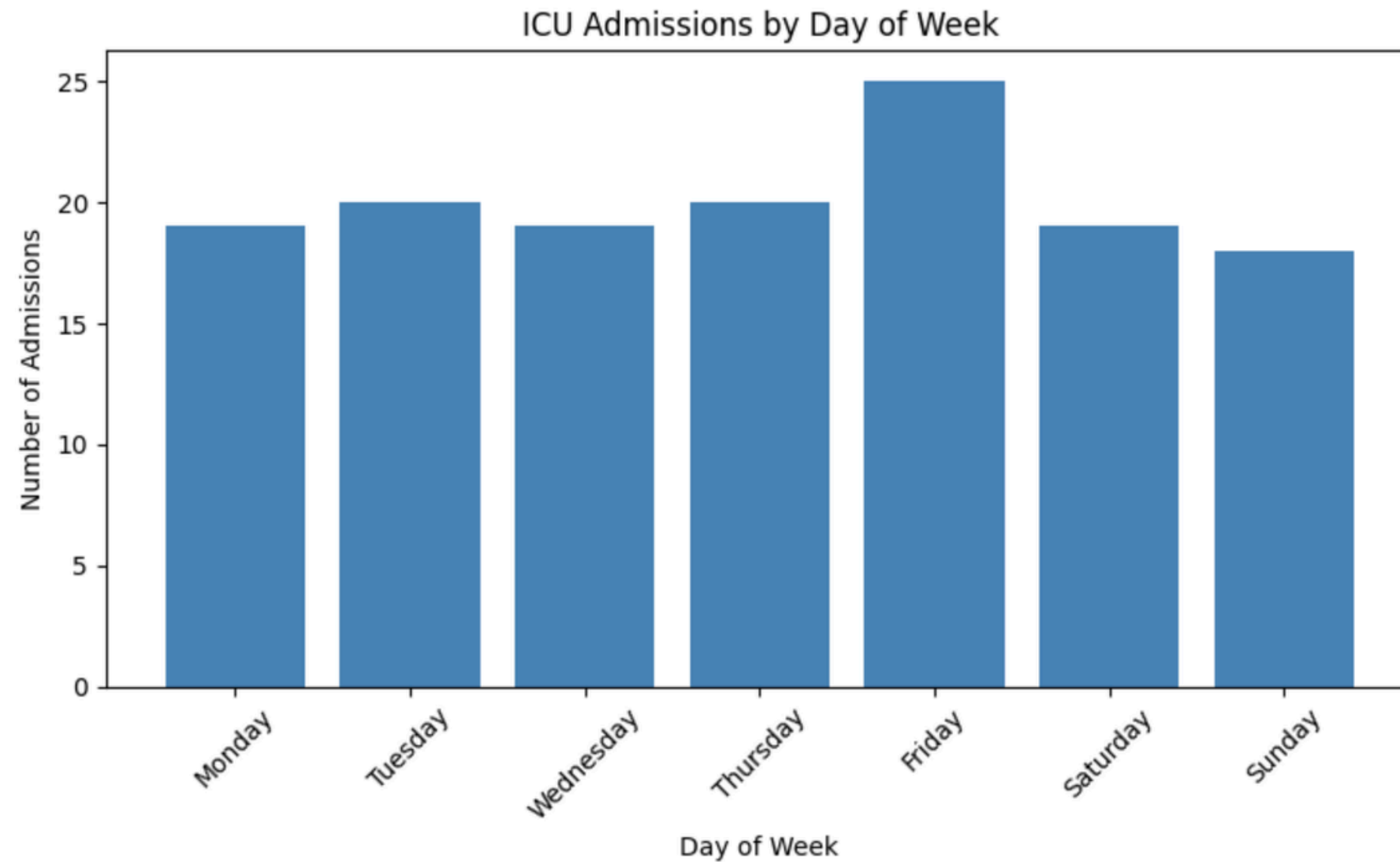
```
icu_category
Neurological    5.510831
Medical         4.123376
Cardiac         3.317627
Surgical        3.268301
Name: los, dtype: float64
```

Total ICU Days by ICU Specialty Category

```
icu_category
Medical         214.415556
Surgical        147.073542
Cardiac         126.069815
Neurological     27.554155
Name: los, dtype: float64
```

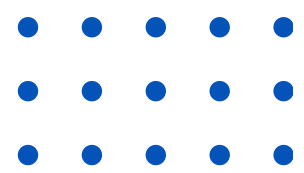


Admissions Patterns by Day of Week



- Admissions peak on Friday
- Lowest admissions on Sunday
- Majority on Weekdays
- Caused by elective procedures

Staffing Recommendations Based on Shift Patterns



Evening shift has
highest average
LOS

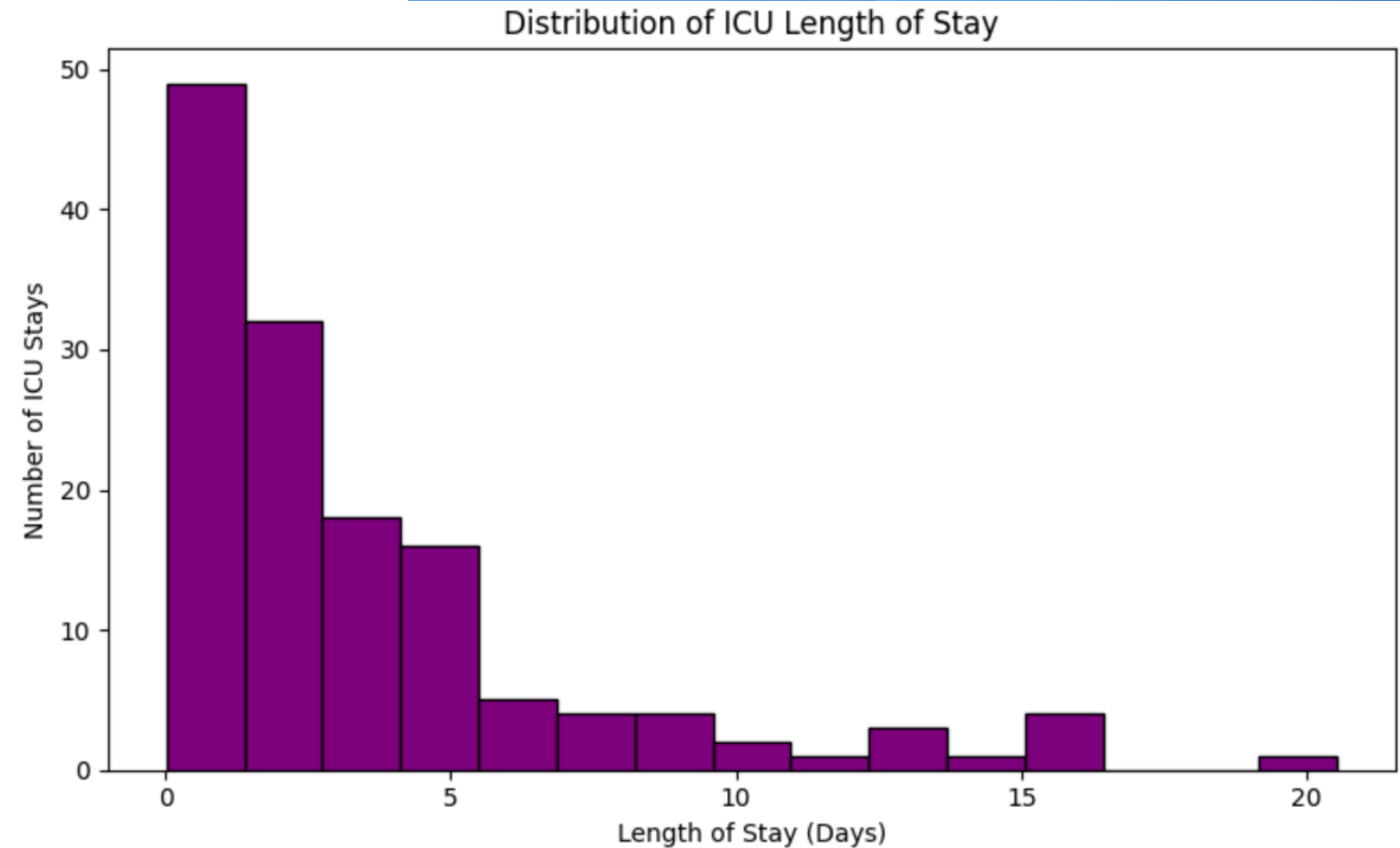
Evening
admissions may
involve higher
complexity

Weekday
evenings
show highest
volume

Recommended:

- Increased coverage evenings
- Stronger admission coordination
- Resource overlap during shift transitions

Length of Stay Distribution



✓ Right-skewed

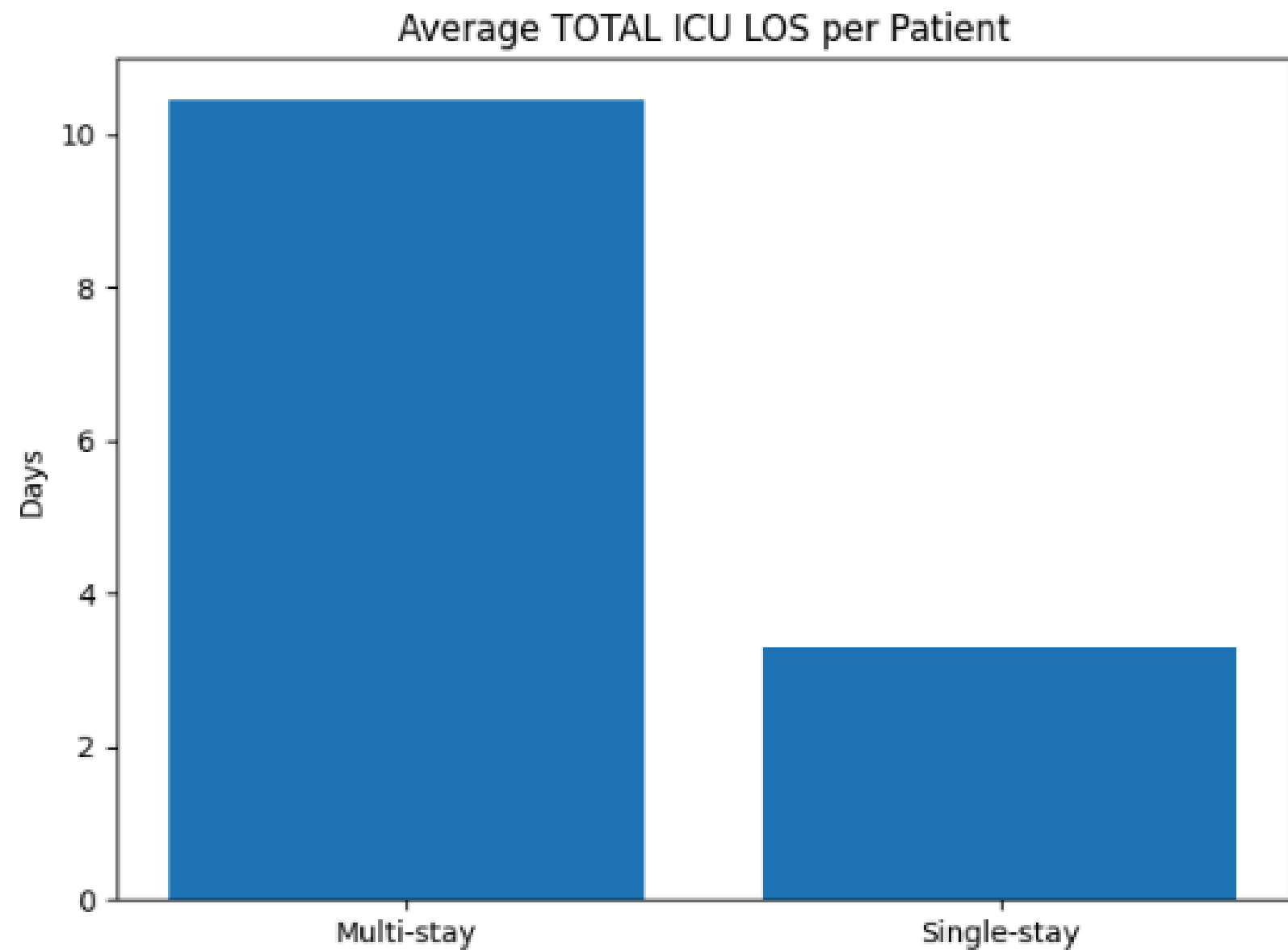
✓ 26% of patients have multiple ICU stays

✓ Long-staying patients reduce bed turnover

✓ Capacity planning must account for LOS variability

Repeat Patients Drive Disproportionate Demand (Total LOS per patient)

- **26%** of patients have 2+ ICU stays
- Those repeat patients account for **47.1%** of ICU stays
- Avg TOTAL ICU LOS per patient:
 - Multi-stay: **10.46** days
 - Single-stay: **3.28** days



✓ A small repeat-patient subgroup drives sustained bed demand, reducing turnover.

Capacity Planning Based on ICU Utilization

Which ICU drives bed demand?

- Medical ICU → 214 total ICU days (largest cumulative occupancy)
- Surgical → 147 ICU days
- Cardiac → 126 ICU days
- Neurological → 27 ICU days

What this tells us:

- Medical ICU represents the greatest sustained bed demand

What this tells us:

- Neurological has longer stays but lower volume

What this tells us:

- ICU specialty is a stronger driver of LOS than timing

Operational Implication:

- Bed allocation should prioritize Medical ICU capacity first.

Patient Flow & Expansion Strategy

✓ Transfer Patterns:

- 8% of stays involved ICU transfers
- Most common pathways:
 - CCU → MICU
 - SICU → MICU
- Medical ICU often acts as receiving unit

✓ What this means:

- Medical ICU functions as an operational hub
- Bottlenecks likely occur when MICU beds are full
- Medical ICU often acts as receiving unit

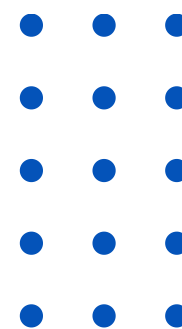
✓ Expansion Recommendation :

If adding beds:

1. Expand Medical ICU first
2. Evaluate Neurological for specialized long-stay capacity
3. Monitor Cardiac & Surgical for surge planning

✓ Bottom Line:

ICU specialty type is primary driver of resource burden.



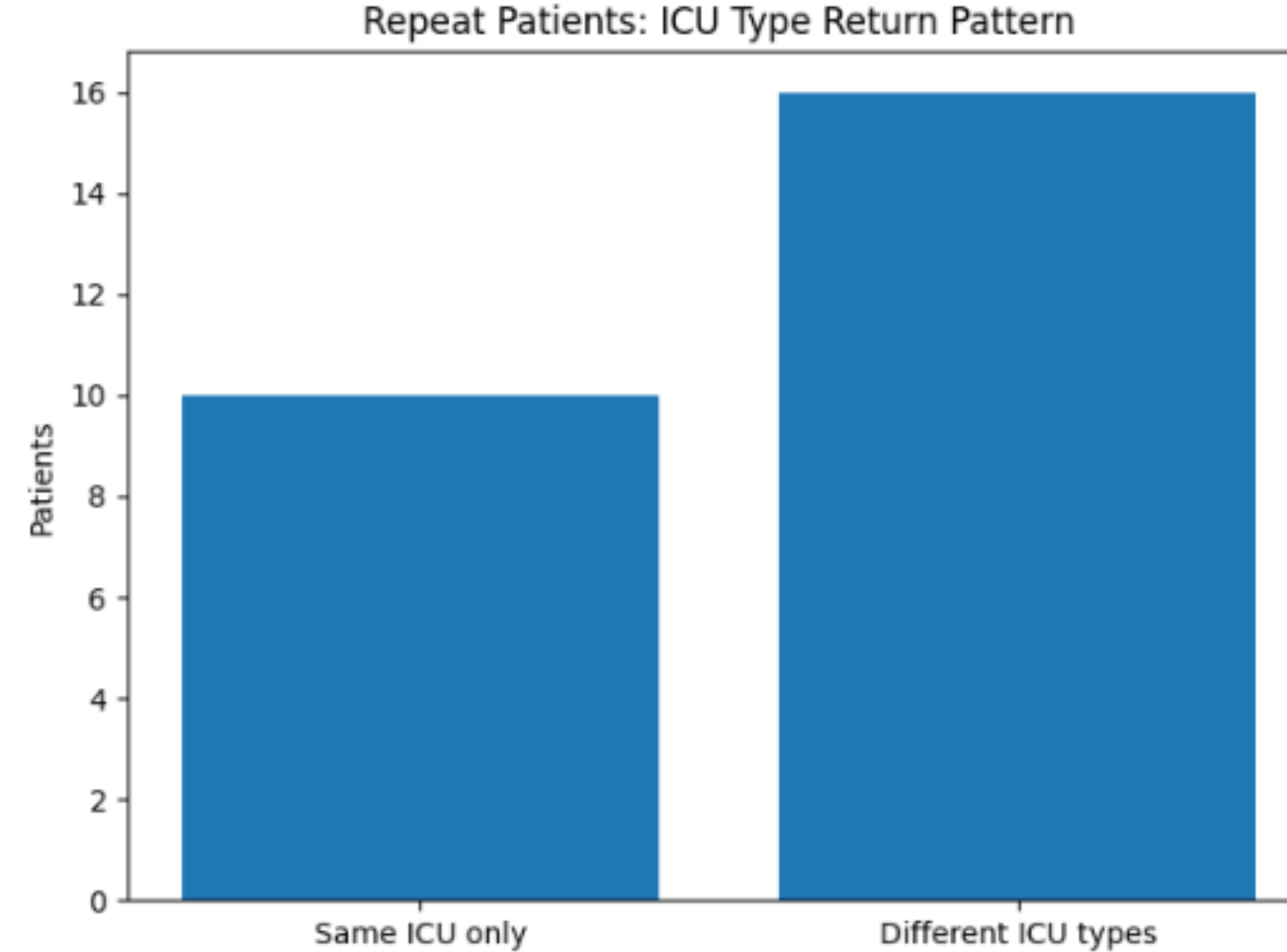
Transfers + Return Patterns (Same vs Different ICU types)

✓ Transfers (all stays)

- Transfers: 11 / 140 (7.9%)
- Most stays remain in the same ICU type, so transfers are uncommon

✓ Repeat patient return pattern (n=26)

- Same ICU only: 10
- Different ICU types: 16



✓ Top transfer pathways

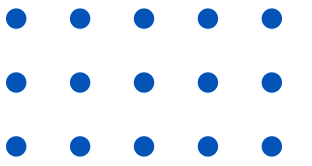
- CCU → MICU: 2
- SICU → MICU: 2
- Others occur 1 time each

✓ Bottom Line:

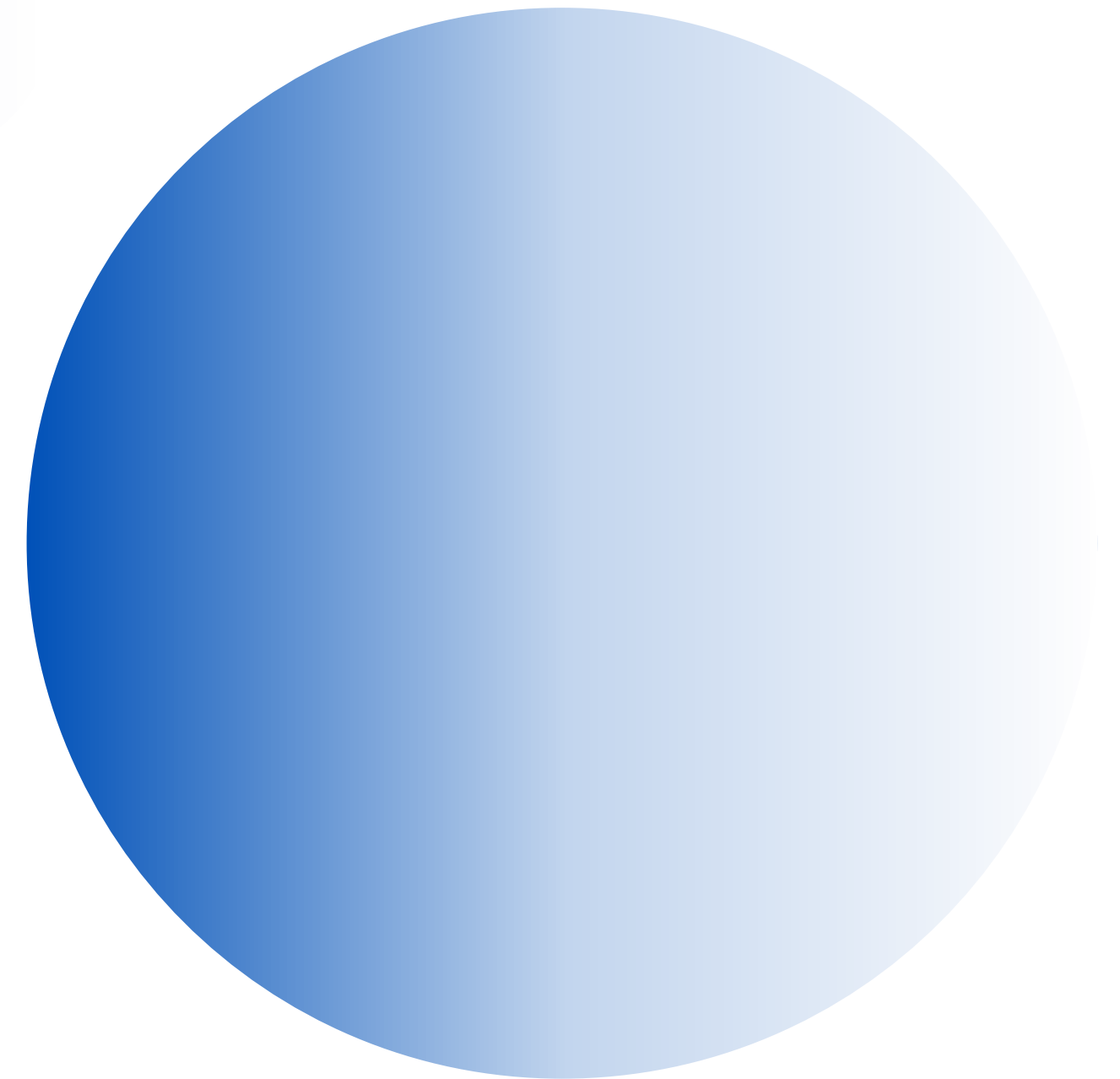
Planning must account for transfers + repeat cross-unit patients, not just LOS.



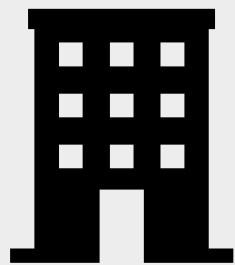
Next Steps



1. Expand the data set
 - Include full MIMIC-IV population
 - Analyze multiple years for seasonal trends
2. Integrate Clinical Severity & Outcomes
 - Add diagnosis codes
 - Include severity scoring
3. Develop Predictive Capacity Models
 - Build LOS prediction model by ICU specialty
 - Forecast bed demand by unit



Expansion Recommendations



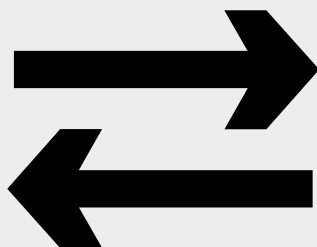
Expand Medical ICU Capacity First

214 total ICU days - largest sustained demand



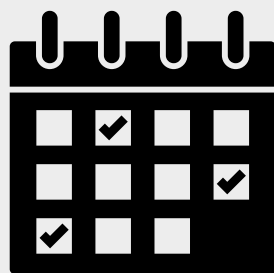
Specialized Staffing by Unit Type

Neurological needs long-term protocols, Surgical needs high-throughput



Optimize Transfer Pathways

Medical ICU acts as a receiving hub - reduce bottlenecks



Weekend Capacity Planning

Friday admission spike requires adjusted staffing levels



Thank You

Any Questions?